Machine Learning Regressin Report

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1 Report Task 1: Model Summaries

1.1 Linear Regression Results

Linear regression is used here to model medical charges. Figure 1 shows the predicted–actual scatter plot, illustrating that the model tracks the diagonal reasonably well, with larger errors mainly occurring for high-cost cases.

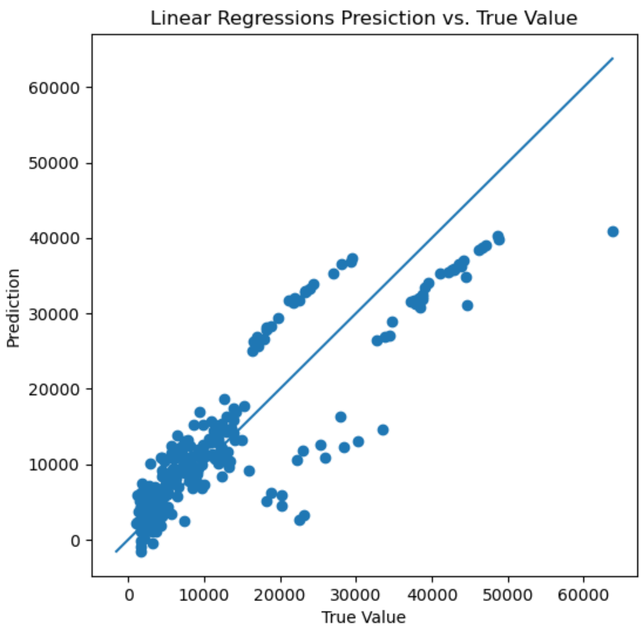

Figure 1: Predicted vs Actual charges for the linear regression model.

Table 1 displays the learned coefficients. Smoking clearly dominates all other predictors, while age and BMI have moderate positive contributions. Regional indicators remain small and consistently negative, suggesting weak geographic variation. Table 2 presents the RMSE and MAE for both training and test sets. The values are close across splits, indicating stable generalisation with no strong signs of overfitting.

Feature	Coefficient
age	3614.975
bmi	2036.228
children	516.890
sex_male	-18.592
smoker_yes	23651.129
region_northwest	-370.677
region_southeast	-657.864
region_southwest	-809.799

Table 1: Learned coefficients.

Metric	Train	Test
RMSE	6105.545	5796.285
MAE	4208.235	4181.194

Table 2: Train vs test performance.

1.2 Neural Network Results

A neural network with three hidden layers (64-32-16 units) and small dropout (p=0.05) was trained to model insurance charges. Figure 2 shows the training and test loss over 300 epochs. Both curves decrease rapidly at the start and then stabilise, remaining close throughout training. This pattern indicates smooth optimisation and no evidence of overfitting, assisted by batch training and dropout regularisation.

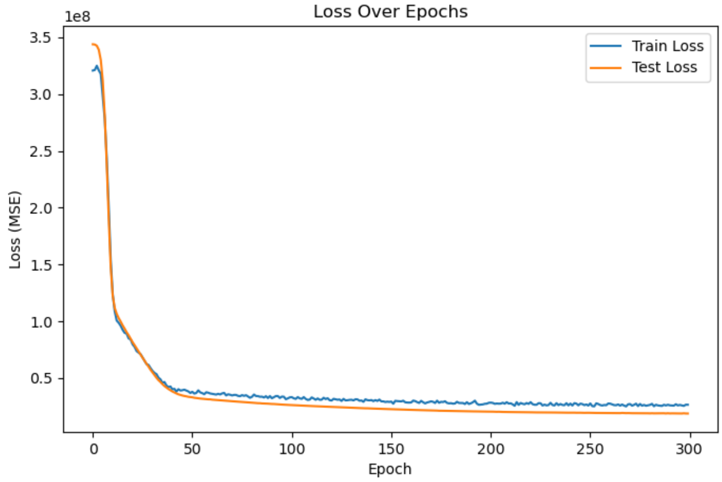

Figure 2: Training and test MSE loss over epochs for the neural network.

Table 3 summarises the final RMSE on the training and test sets. The values are close, with the test RMSE slightly lower, confirming strong generalisation and the effectiveness of mild regularisation.

Metric	Train	Test
RMSE	4675.721	4331.258

Table 3: Training and testing RMSE for the neural network model.

Figure 3 presents the predicted–actual scatter plot on the test set. The predictions align well with the diagonal, with most points concentrated near the line and larger deviations appearing mainly for high-charge cases where variance in the data is naturally higher.

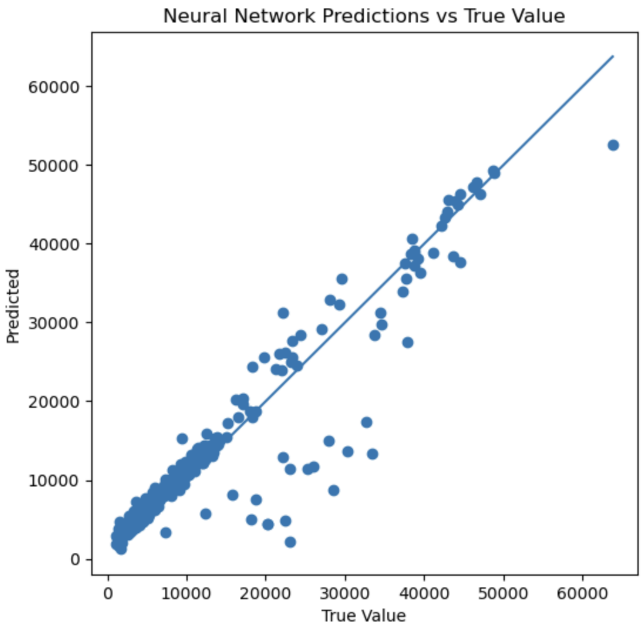

Figure 3: Predicted vs actual charges on the test set for the neural network.

1.3 Bayesian Linear Regression

A Bayesian linear regression model was fitted using PyMC with weakly informative Normal priors over the coefficients and a HalfNormal prior over the noise scale. Posterior samples were obtained using the NUTS sampler (4 chains, 1000 draws per chain after 1000 tuning steps). The posterior predictive performance is also consistent with the earlier linear regression model, with an RMSE of 5796.815 on the test set (almost identical to the linear model's value), indicating that the Bayesian formulation captures the same underlying structure while providing full uncertainty quantification. Table 4 reports the posterior means for all model coefficients and the noise parameter.

Feature	Posterior Mean
Intercept (α)	8966.663
age	3613.007
bmi	2036.319
children	516.262
sex_male	-25.827
smoker_yes	23622.942
region_northwest	-369.810
region_southeast	-658.999
region_southwest	-819.758
Noise (σ)	6135.531

Table 4: Posterior means for all regression coefficients and noise parameter in the Bayesian model.

2 Report Task 2: Model Comparison (Linear Regression vs. Neural Network)

The linear regression model achieved a test RMSE of 5796.285, while the neural network achieved a lower test RMSE of 4331.258. This shows that the neural network fits the data more accurately on unseen examples and captures relationships that the linear model cannot represent. If the underlying relationship were genuinely linear or close to linear, a simpler model such as linear regression could outperform a neural network by avoiding unnecessary complexity.

Linear regression is simple, fast, and completely interpretable: each coefficient corresponds directly to the marginal contribution of a feature. The model assumptions are strong (linearity, additivity), and the predictions are limited to relationships that can be expressed as weighted sums of the inputs. Despite this simplicity, its performance is stable, and the training time is negligible.

The neural network, by contrast, is a more complex model with multiple hidden layers and nonlinear activations. This flexibility allows it to learn curved, interaction-dependent patterns in the data. This explains the visible reduction in RMSE. However, this comes at the cost of interpretability: the meaning of individual weights is not accessible, and the model behaves as a black box.

Computationally, the neural network takes longer to train because gradient-based optimization must update thousands of parameters over hundreds of epochs. The model also has a higher inherent risk of overfitting, since its capacity is far larger than that of linear regression. In this case, smooth training curves and the use of dropout regularisation helped prevent overfitting, keeping the train/test losses close.

Overall, the neural network provides better predictive accuracy due to its ability to model nonlinear structure, but this improvement requires more computation and sacrifices interpretability. Linear regression remains useful as a simple, transparent baseline, while the neural network offers superior performance when accuracy is prioritised.